

ECHS PRE-AUTHORIZATION FRAUD ANALYTICS

Combined Report: Type 1 and Type 2

PROBLEM 14 REPORT | Type 1: 2012–2017 | Type 2: Dec 2021 – Mar 2026

CLASSIFICATION	PERIOD	RECORDS ANALYSED	QUERY SETS
RESTRICTED	2012–2026	~43.5M Claims	Problem 14

IIT KANPUR — Data Analytics & Fraud Intelligence Division

14 May 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

This report covers the two pre-authorization claim types in ECHS that carry a declared cost estimate enabling systematic fraud analysis: Type 1 (pre_auth, discontinued 2012-2017) and Type 2 (unlisted_procedure, active December 2021 to March 2026). After deduplication, Type 1 yields 2,163 unique claims and Type 2 yields 2,25,708 unique claims. Combined excess billing identified: **Rs 1,282 Cr**. Other than this the only candidate found in the analysis that could potentially have a pre-authorization record was the prior approval table which had zero rows potentially meaning that system was never activated.

Type 1 Claims 2,163 (deduped)	Type 2 Claims 225,708 (deduped)	T2 Excess Billed Rs 1273 Cr Above Sanction	Combined Rs 1,282 Cr Excess Billed
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Key Findings at a Glance

Finding	Type	Impact
Bill inflation: Rs 2.93 Cr estimate became Rs 11.87 Cr billed	T1	4.1x inflation
UTI review cut 19% off billed (PAR/SUP/APP) - only partial mitigation	T1	Rs 8.02 Cr final approved
83 breach cases (11.8%) - bill exceeded 125% of estimate	T1	Rs 67.45 L excess
Rubber stamp approval - 0% sanction cut in every Type 2 case	T2	Rs 1,190 Cr sanctioned at estimate
Sanction not enforced as payment cap - UTI paid above sanction	T2	Rs 1,273 Cr excess billed
Anomalous Rs 2L clustering at non-threshold amount	T2	Unknown reason
Top 12 hospitals: avg bill/sanction ratio 7x to 16x	T2	Needs to be reviewed

PRE-AUTHORIZATION SCOPE: TYPE 1 AND TYPE 2

Total ECHS claims across all time and all claim types: 43,510,015. Pre-auth linked claims (all types combined, all years): 363,689 - representing **0.84%** of the entire ECHS universe. Type 2 alone (225,708 claims, **0.52%** of total ECHS volume) accounts for **Rs 2136.73 Cr** in gross billing.

ECHS - UTI PAYMENT PIPELINE: INSTITUTIONAL ROLES AND ASSUMPTIONS

The assumption based on which analysis is done is that the pipeline for both pre-auth types follows four stages:

- (1) Hospital submits pre-auth request with cost estimate to ECHS.
- (2) ECHS reviews and issues a decision. For Type 1: binary approve/reject with no rupee ceiling set. For Type 2: sanction always equals the declared estimate (verified: ratio = 1.000 across all 2,13,074 approved rows).
- (3) Hospital treats the patient and submits the bill directly to UTI.
- (4) UTI (the third-party administrator) runs a three-stage expense review (PAR, SUP, APP) and makes payment.

DEDUPLICATION LOGIC: WHY IT IS NEEDED AND HOW IT DIFFERS BY TYPE

Both source tables contain more rows than unique claims. Applying financial analysis without deduplication would either overcount (raw SUM) or undercount (take any single row for multi-procedure claims). The correct strategy differs because the reason for multiple rows differs by type.

Dimension	Type 1 (pre_auth)	Type 2 (unlisted_procedure)
Why multiple rows	Same claim re-submitted with revised details	74.2% different procedures; 25.8% same procedure revised
Correct strategy	Keep LATEST row per pa_intimation_id (by pa_date)	Latest per (claim, procedure text) then SUM across distinct procedures
Rows collapsed	786 earlier submissions removed	15,415 revised rows collapsed (Rs 105 Cr correction vs raw SUM)

TYPE 1 ANALYSIS - pre_auth (April 2012 to October 2017)

After deduplication: 2,163 unique claims. The pre_auth workflow recorded the approval decision for each proposed treatment. Despite **97.3%** of all submissions being rejected (pa_approved = N), the downstream payment system operated independently of the pre-auth decision. The system was discontinued in 2018.

Unique Claims 2,163 (after dedup)	Duplicate Rows 786 Removed	Claims Settled 90.8% and Paid	Total Hospital Rs 11.87 Cr Billed
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Fraud Finding 1 - Approval Gate Had No Effect on Payment

The pre_auth system issued three outcomes: Y (approved), N (rejected), X (cancelled). At row level, **97.3%** of all 2,949 submissions were recorded as pa_approved = N. Settlement records show that rejected claims were settled and paid at nearly the same rate as approved claims. The payment pipeline (UTI processing) operated independently of the ECHS pre-auth decision.

ECHS Decision	Unique Claims	Claims Settled	% Settled	Estimate	Net Paid
Rejected (pa_approved = N)	2,097	1,920	91.6%	Rs 2.90 Cr	Rs 8.24 Cr
Cancelled (pa_approved = X)	13	8	61.5%	Rs 0.00 Cr	Rs 0.11 Cr
Approved (pa_approved = Y)	53	35	66.0%	Rs 0.03 Cr	Rs 0.23 Cr

Key Implication: ECHS approval status has no bearing on whether UTI paid

For Type 1, pa_approved = N claims were paid at the same rate as pa_approved = Y claims. ECHS and UTI operated as independent systems with no data link between the pre-auth decision and the payment trigger.

Fraud Finding 2 - Extreme Bill Inflation After ECHS Decision

For the 767 Type 1 claims that submitted a cost estimate at the ECBS pre-auth stage (38.3% of all deduplicated claims), the total estimate was Rs 2.93 Cr. After ECBS made its decision (97.3% rejected), hospitals billed Rs 11.87 Cr to UTI - a 4.0x inflation over the declared estimate. UTI then reviewed and cut the bill, approving Rs 9.61 Cr (PAR/SUP/APP stages: -19.0% of billed), with net paid at Rs 8.58 Cr.

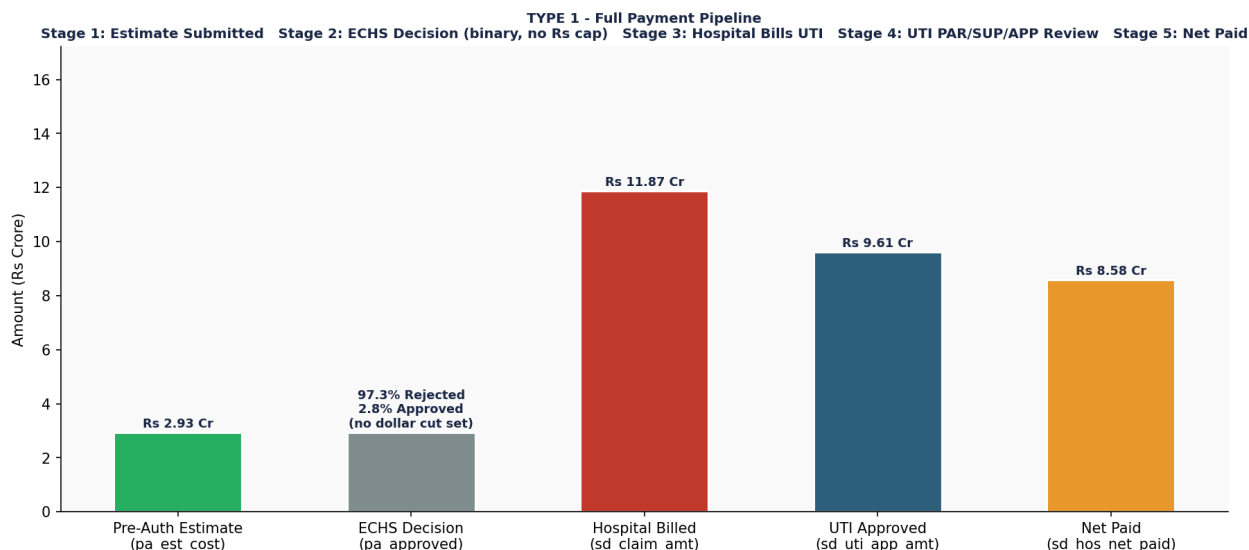


Figure 4 - TYPE 1: Full 5-stage payment pipeline. Stage 1: estimate submitted. Stage 2: ECBS binary decision (no Rs cap). Stage 3: hospital bills UTI. Stage 4: UTI PAR/SUP/APP review. Stage 5: net paid.

Type 1 - UTI Expense Review: PAR / SUP / APP

For 1,683 of the 2,163 Type 1 claims, UTI ran a three-stage expense review: PAR (Primary Authorisation Review), SUP (Supplementary Review), APP (Final Approval). This is Stage 4 of the pipeline - after ECBS decision and hospital billing. Hospital claimed **Rs 9.90 Cr**. After all three UTI stages, **Rs 8.02 Cr** was approved - a **19.0%** reduction. The SUP stage made the heaviest cut.

UTI Review Stage	Amount (Rs Cr)	Stage Cut	Notes
Hospital Claimed (Stage 4 input)	Rs 9.90 Cr	Baseline	Submitted after ECBS decision and hospital billing
After PAR	Rs 9.74 Cr	-1.6%	Primary review - smallest cut
After SUP	Rs 8.10 Cr	-16.9%	Supplementary review - heaviest cut
After APP (final)	Rs 8.02 Cr	-1.0%	Final UTI approval
Total UTI cut		-19.0%	vs. hospital claimed amount

The UTI expense review provided meaningful financial scrutiny, cutting **19.0%** off the claimed amount. However, this is Stage 4 of the pipeline - it occurs after both the ECBS decision and the hospital bill are already committed. Compare: Type 2 has **0%** cut at the ECBS sanction stage, and UTI has no mechanism to enforce the sanction ceiling when the bill arrives.

Type 1 - Full Financial Pipeline: All 5 Stages vs Pre-Auth Estimate

Pipeline Stage	Amount (Rs Cr)	vs Estimate	vs Hospital Billed	Notes
Stage 1: Pre-Auth Estimate	Rs 2.93 Cr	Baseline	-	767 claims with estimate > 0
Stage 2: ECHS Decision	No Rs amount set	ECHS issues Y/N/X - no ceiling	-	97.3% rejected; payment occurred regardless
Stage 3: Hospital Billed	Rs 11.87 Cr	+305%	Baseline	4.0x estimate; submitted directly to UTI
Stage 4: UTI Approved	Rs 9.61 Cr	+228%	-19.0%	After PAR/SUP/APP; UTI cut applied here
Stage 5: Net Paid	Rs 8.58 Cr	+192%	-27.7%	Final transfer to hospital

TYPE 1 - BREACH ANALYSIS (Deduped, Estimate >= Rs 100)

Of 703 analysable Type 1 claims (pa_est_cost >= Rs 100, with a hospital bill), 83 (11.8%) showed a hospital bill more than 25% above the pre-auth estimate - classified as breach cases. Total excess billed above estimate: Rs 67.45 L. All numbers are computed from the deduplicated dataset.

Analysable Claims 703 est >= Rs 100, has bill	Breach Cases 83 Bill > 1.25x estimate (11.8%)	Excess Billed Rs 67.45 L vs pre-auth estimate	Net Paid Rs 64.22 L (Breach Set)
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Breach Cases - Financial Comparison at All Pipeline Stages

Pipeline Stage	Amount (Rs L)	vs Pre-Auth Estimate	vs Hospital Billed
Pre-Auth Estimate (Stage 1)	Rs 23.82 L	Baseline	-
ECHS Decision (Stage 2)	No Rs amount	Binary Y/N, no ceiling	-
Hospital Billed (Stage 3)	Rs 91.27 L	+283.1%	Baseline
UTI Approved (Stage 4)	Rs 71.67 L	+200.9%	-21.5%
Net Paid (Stage 5)	Rs 64.22 L	+169.6%	-29.6%
Excess (Billed - Estimate)	Rs 67.45 L	Excess above baseline	-

Year-wise Breakdown - All Analysable Claims

FY	Claims	Estimate	Billed	UTI Approved	Net Paid	Breach %
2012-13	108	Rs 65.88 L	Rs 67.50 L	Rs 54.72 L	Rs 48.85 L	16.7%
2013-14	76	Rs 41.53 L	Rs 39.95 L	Rs 30.87 L	Rs 27.50 L	17.1%

FY	Claims	Estimate	Billed	UTI Approved	Net Paid	Breach %
2014-15	291	Rs 83.56 L	Rs 79.03 L	Rs 63.21 L	Rs 56.23 L	6.5%
2015-16	178	Rs 67.06 L	Rs 63.24 L	Rs 49.97 L	Rs 44.49 L	9.6%
2016-17	31	Rs 23.50 L	Rs 25.19 L	Rs 22.67 L	Rs 20.17 L	19.4%
2017-18	19	Rs 6.49 L	Rs 18.38 L	Rs 13.98 L	Rs 12.45 L	52.6%

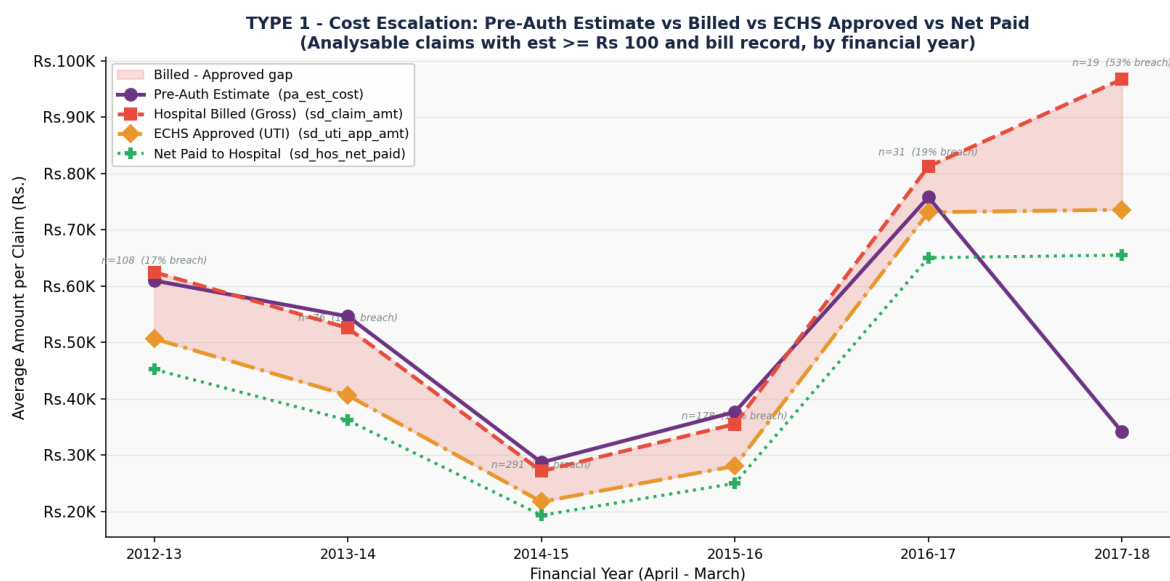


Figure 6 - TYPE 1: Average per-claim cost escalation by financial year (703 analysable claims, est >= Rs 100). Lines show all 5 pipeline stages. Breach % = fraction of claims where bill exceeded **125%** of estimate.

TYPE 1 - FLAGGED CASES (Four Fraud Patterns, Deduped)

Four distinct fraud patterns identified in the Type 1 breach dataset. All case counts and amounts are from the deduplicated dataset.

Flag 1 - Systematic Overbilling by Single Hospital (ID: prayaghos, name: PRAYAG HOSPITAL & RESEARCH CENTRE PVT LTD)

Hospital prayaghos: 14 breach cases, **Rs 23.80 L** excess (estimate **Rs 5.16 L**, billed **Rs 28.96 L**, **461%** overrun). Drives **35.3%** of all Type 1 breach excess.

PA ID	Date	Diagnosis	Estimate	Billed	UTI Approved	Net Paid	Breach %
1482934	2014-05-27	fracture pelvis with fracture dislocated	Rs 62,000	Rs 833,631	Rs 639,223	Rs 575,025	1244.6%
1053748	2013-12-02	COPD WITH DM TYPE2 WITH PLEURAL EFFUSIO	Rs 30,000	Rs 408,283	Rs 213,964	Rs 192,293	1260.9%

PA ID	Date	Diagnosis	Estimate	Billed	UTI Approved	Net Paid	Breach %
1130400	2013-12-26	COPD , DM TYPE 2, PULM KOCH'S WITH SEIZU	Rs 9,990	Rs 287,282	Rs 245,852	Rs 220,993	2775.7%
456506	2013-03-23	ACUTE PIVD WITH LUMBER RADICULOPATHY WIT	Rs 120,000	Rs 232,391	Rs 175,697	Rs 157,858	93.7%
332200	2012-12-19	B/L CARPAL TURNAL SYNDROME POST OPERATIV	Rs 18,000	Rs 183,181	Rs 112,448	Rs 100,934	917.7%
356400	2012-12-22	CVA LT. SIDE HEMIPARESIS WITH HYPOPROTEN	Rs 125,000	Rs 178,990	Rs 141,150	Rs 126,766	43.2%
723309	2013-07-16	I3 I4 pibd with paraparesis lt. leg with	Rs 16,000	Rs 171,425	Rs 166,403	Rs 149,494	971.4%
294196	2013-01-20	ACUTE RD WITH FEVER WITH METASTATIC WITH	Rs 20,000	Rs 156,995	Rs 113,671	Rs 102,035	685.0%
339717	2012-12-29	BED SORE WITH PARASIS BOTH LOWER LIMB WI	Rs 15,000	Rs 131,157	Rs 114,552	Rs 102,828	774.4%
335257	2012-12-20	ACUTE ABDOMEN WITH BPH WITH FEVER	Rs 35,000	Rs 86,553	Rs 62,996	Rs 56,427	147.3%
1126108	2013-12-16	FRACTURE OLEARANEN WITH FRACTURE WITH	Rs 8,250	Rs 66,983	Rs 17,522	Rs 15,191	711.9%
356395	2012-12-22	ACUTE CHOLECYSTITIES WITH CHOLELITHIASIS	Rs 12,000	Rs 66,263	Rs 12,000	Rs 10,531	452.2%
356389	2012-12-27	manorrhagia with pid with acute abdomen c	Rs 35,000	Rs 61,970	Rs 25,757	Rs 22,912	77.1%
560399	2013-05-02	PARA UMBLICAL HERNIA	Rs 9,900	Rs 31,355	Rs 31,019	Rs 27,648	216.7%

Flag 2 - Formally Approved (pa_approved=Y) Yet Bill Exceeded Estimate

3 claims with pa_approved = Y still had hospital bills exceed the pre-auth estimate. These are the rare cases where ECHS approved AND the hospital overbilled.

PA ID	Date	Approver	Hospital	Diagnosis	Estimate	Billed	Net Paid	Breach %
208340	2012-10-12	app.tv m	mahe.p wr	DIABETIC FOOT GANGRENE	Rs 10,000	Rs 21,244	Rs 10,343	112.4%
84991	2012-07-09	app.pu n	nan	Aspiration Pneumonia+ Parkinso	Rs 30,700	Rs 126,834	Rs 85,817	313.1%
92136	2012-06-26	app.pu n	nan	ABSCESS LEFT ELBOW+FOREARM WIT	Rs 7,500	Rs 107,500	Rs 89,731	1333.3%

Flag 3 - Zero-Estimate Claims Paid (pa_est_cost=0, Net Paid > 0)

767 claims had pa_est_cost = 0 at Stage 1 (no declared cost baseline) yet received net payment of Rs 284.37 L at Stage 5. With no estimate, UTI had no benchmark against which to apply the PAR/SUP/APP review.

Hospital	Zero-Est Claims	Total Billed	Net Paid
EMLC1718	50	Rs 6.88 L	Rs 6.05 L
getwell	41	Rs 1.70 L	Rs 1.45 L
PRAYAG HOSPITAL & RESEARCH CENTRE PVT LTD	34	Rs 13.01 L	Rs 9.49 L
Nanda Hospital - UNA	31	Rs 6.30 L	Rs 5.24 L
GRECIAN SUPER - SPECIALITY HOSPITAL	30	Rs 11.25 L	Rs 9.69 L
Shree Balaji Superspeciality Hospital	22	Rs 8.92 L	Rs 5.89 L
ANASUYA INSTITUTE OF MEDICAL SCIENCES	20	Rs 3.88 L	Rs 3.20 L
Fortis Hospital - Mohali	19	Rs 37.86 L	Rs 26.97 L

TYPE 2 ANALYSIS - unlisted_procedure (December 2021 to March 2026)

After deduplication: 225,708 unique claims across December 2021 to March 2026. The pipeline for Type 2 has the same 5 stages as Type 1, but the ECHS sanction (Stage 2) always equals the declared estimate (verified: ratio = 1.000 across all 2,13,074 approved rows).

Type 2 Pre-Auth Penetration Rate

Out of 23,969,480 total ECHS hospital admissions (Dec 2021 to Mar 2026), 225,876 (0.94%) triggered a Type 2 pre-auth. The penetration rate grew nearly 19x from 0.06% in 2021 to 1.15% in 2025.

Year	Total ECHS Admissions	Type 2 Pre-Auths Filed	Penetration Rate
2021	286,506	167	0.06%
2022	3,915,607	20,883	0.53%
2023	4,847,900	46,403	0.96%
2024	5,982,326	56,513	0.94%
2025	7,185,116	82,291	1.15%
2026	1,752,025	19,619	1.12%

Pattern 1 - Pre-Auth Rubber Stamping: Zero Financial Scrutiny at Stage 2

When a Type 2 pre-auth is submitted, ECHS sanctions it at exactly the declared estimate - without any reduction in any case. This is universal: across all 2,13,074 approved procedure rows, the sanctioned amount equals the declared estimate (ratio = 1.000). The polyclinic reviewer approves at the hospital's own price or rejects entirely. There is no partial-sanction mechanism, no CGHS benchmarking, and no cost negotiation.

Pattern 2 - Bill Inflation After ECHS Sanction: Stage 3 Exceeds Stage 2

Pipeline Stage	Amount (Rs Cr)	vs ECHS Sanction (Stage 2)	vs Hospital Billed (Stage 3)
Stage 1+2: ECHS Sanction (= Estimate)	Rs 1036.76 Cr	Baseline	-
Stage 3: Hospital Billed	Rs 2136.73 Cr	+106.1%	Baseline
Stage 4: UTI Approved	Rs 1556.80 Cr	+50.2%	-27.1%
Stage 5: Net Paid	Rs 1392.52 Cr	+34.3%	-34.8%
Excess billed above sanction	Rs 1273.23 Cr	71.4% of claims exceeded sanction	-

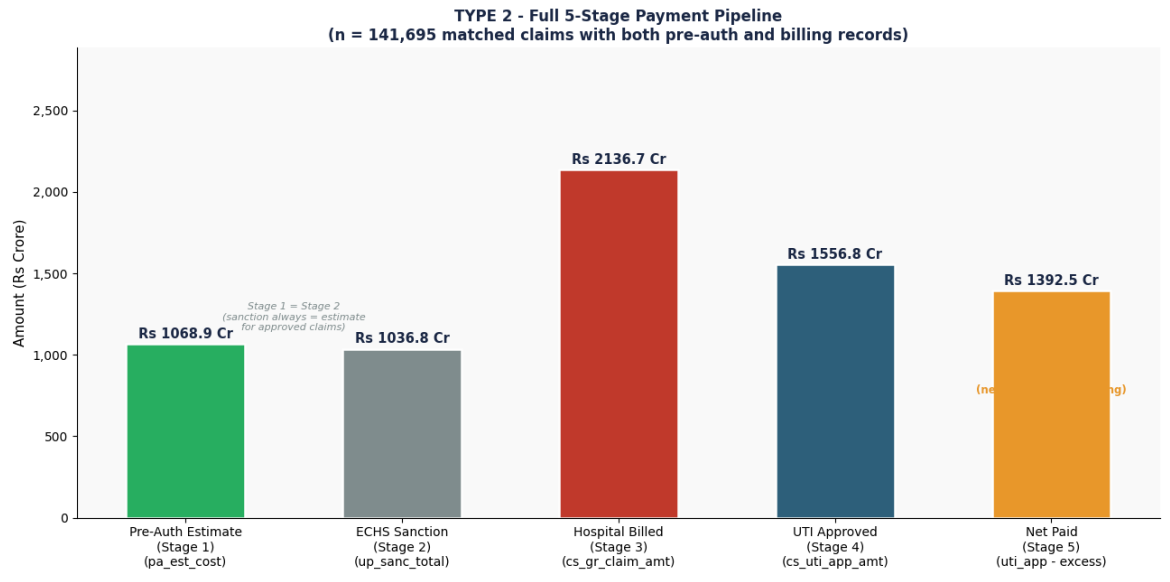


Figure 8 - TYPE 2: Pre-auth sanction vs gross bill vs total paid.

TYPE 2 - Cost Escalation by Financial Year (Matched Claims, n=1,41,695)

The table and chart below show the average per-claim financial pipeline (Stages 2-5) for each financial year, using matched claims only. Avg Bill/Sanction ratio increased from 1.74x in 2021-22 to 2.10x in 2024-25.

FY	Claims	Avg Sanction (Stage 2)	Avg Billed (Stage 3)	Avg UTI Approved (Stage 4)	Avg Net Paid (Stage 5)	Avg Bill/ Sanction
2021-22	1,172	Rs 76.5K	Rs 133.1K	Rs 121.9K	Rs 114.5K	1.74x
2022-23	18,223	Rs 65.8K	Rs 126.5K	Rs 114.9K	Rs 106.2K	1.92x

FY	Claims	Avg Sanction (Stage 2)	Avg Billed (Stage 3)	Avg UTI Approved (Stage 4)	Avg Net Paid (Stage 5)	Avg Bill/ Sanction
2023-24	30,874	Rs 68.2K	Rs 138.8K	Rs 117.9K	Rs 106.0K	2.04x
2024-25	38,823	Rs 74.2K	Rs 156.1K	Rs 126.4K	Rs 112.8K	2.10x
2025-26	52,603	Rs 77.9K	Rs 162.8K	Rs 91.0K	Rs 79.9K	2.09x

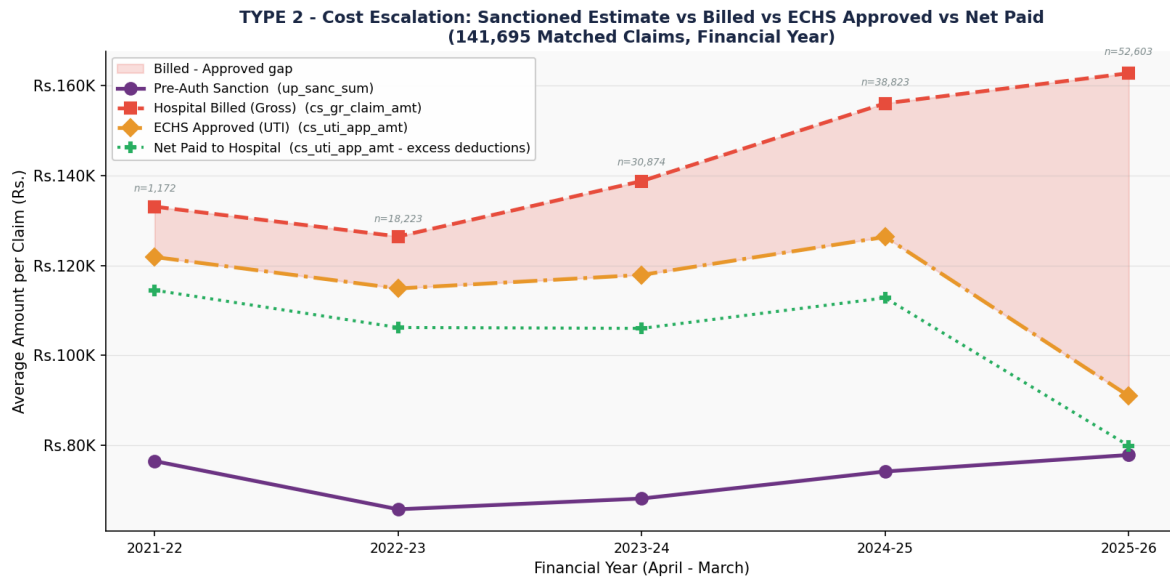
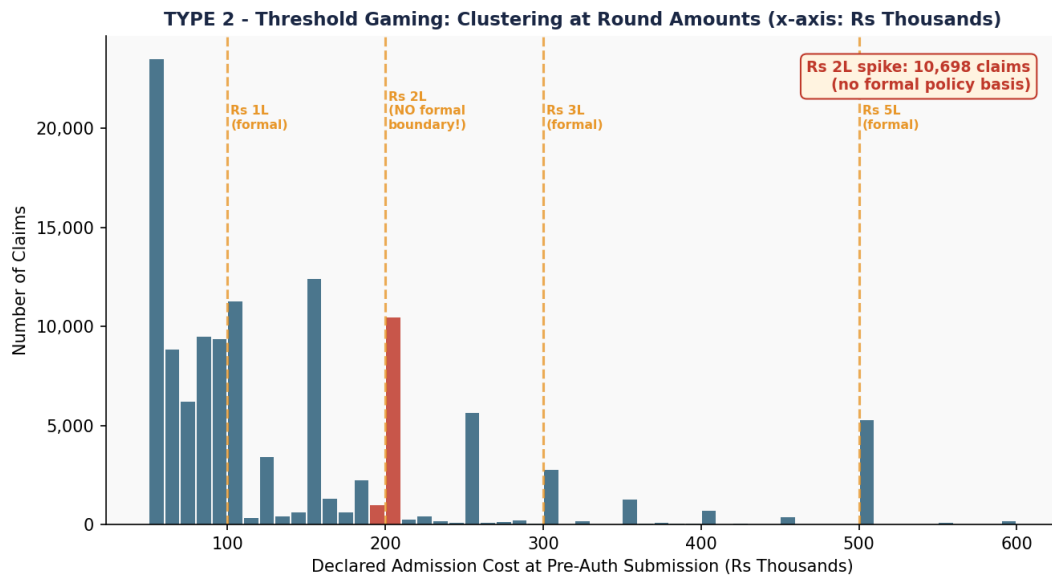


Figure 9 - TYPE 2: Average per-claim cost escalation by financial year. Lines: Stage 2 (sanction), Stage 3 (billed), Stage 4 (UTI approved), Stage 5 (net paid). Bars: avg bill/sanction ratio.

Pattern 3 - Anomalous Clustering at Rs 2 Lakh

A histogram of declared pre-auth estimates shows strong anomalous clustering at Rs 2,00,000 - a value with no formal tier boundary in the ECHS reviewer authority structure. Rs 1,00,000 and Rs 3,00,000 are formal thresholds; Rs 2,00,000 is not. The spike at Rs 2L is stronger than at any formal threshold, indicating hospitals are deliberately targeting this amount to stay within polyclinic sanctioning authority while maximising the declared estimate.



Top 12 Hospitals by Excess Billed Above Pre-Auth Sanction

Hospital ID	Claims	Gross Billed (Rs Cr)	UTI Paid (Rs Cr)	Excess Billed (Rs Cr)	Disallow Rate	Avg Bill / Sanction
PARK HOSPITAL (A UNIT OF UMKAL HEALTHCARE PVT LTD)	1,324	Rs 68.11	Rs 45.85	Rs 56.70	32.7%	10.6x
MEDANTA THE MEDICITY - GURGAON	2,016	Rs 71.03	Rs 42.46	Rs 46.15	40.2%	12.8x
VENKATESHWAR HOSPITAL - DWARKA	1,547	Rs 53.51	Rs 33.77	Rs 44.93	36.9%	16.0x
PARK HOSPITAL (A UNIT OF PARK MEDICENTRES & INSTITUTIONS PVT LTD) - GURGAON	961	Rs 54.62	Rs 33.13	Rs 42.26	39.3%	13.7x
HUMAN CARE MEDICAL CHARITABLE TRUST	1,538	Rs 41.32	Rs 27.80	Rs 32.95	32.7%	11.7x
THE SIGNATURE HOSPITAL (A UNIT OF PARK MEDICITY)	827	Rs 35.05	Rs 25.72	Rs 30.63	26.6%	11.8x
METRO HEART INSTITUTE - FARIDABAD	945	Rs 40.40	Rs 20.28	Rs 28.04	49.8%	7.2x
MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED	905	Rs 30.80	Rs 19.88	Rs 24.93	35.5%	12.7x
SARVODAYA HOSPITAL & RESEARCH CENTRE - FARIDABAD	1,493	Rs 33.23	Rs 25.25	Rs 24.49	24.0%	10.3x
kimsechs Krishna Institute of Medical Sciences Ltd - Secunderabad	765	Rs 16.15	Rs 13.56	Rs 12.58	16.1%	10.3x

Hospital ID	Claims	Gross Billed (Rs Cr)	UTI Paid (Rs Cr)	Excess Billed (Rs Cr)	Disallow Rate	Avg Bill / Sanction
MAX SUPER SPECIALITY HOSPITAL - SHALIMAR BAGH	504	Rs 15.60	Rs 11.34	Rs 11.94	27.3%	11.7x
GRAPHIC ERA INSTITUTE OF MEDICAL SCIENCES	381	Rs 14.84	Rs 6.55	Rs 11.24	55.8%	11.1x

How to Interpret the Avg Bill/Sanction Ratio for Hospitals

The ratio compares total hospitalization bill (cs_gr_claim_amt) against the unlisted procedure pre-auth sanction alone (up_sanc_sum). Since the total bill includes standard charges (room, nursing, medicines) beyond the pre-auth scope, ratios above 1.0x are structurally expected. What is anomalous is when specific hospitals show ratios of 10x to 16x consistently across hundreds of claims: this indicates hospitals are using small pre-auth sanctions as a foot-in-the-door approval to then submit a total bill many multiples higher. The pre-auth is being used as an entry token rather than a genuine cost ceiling.